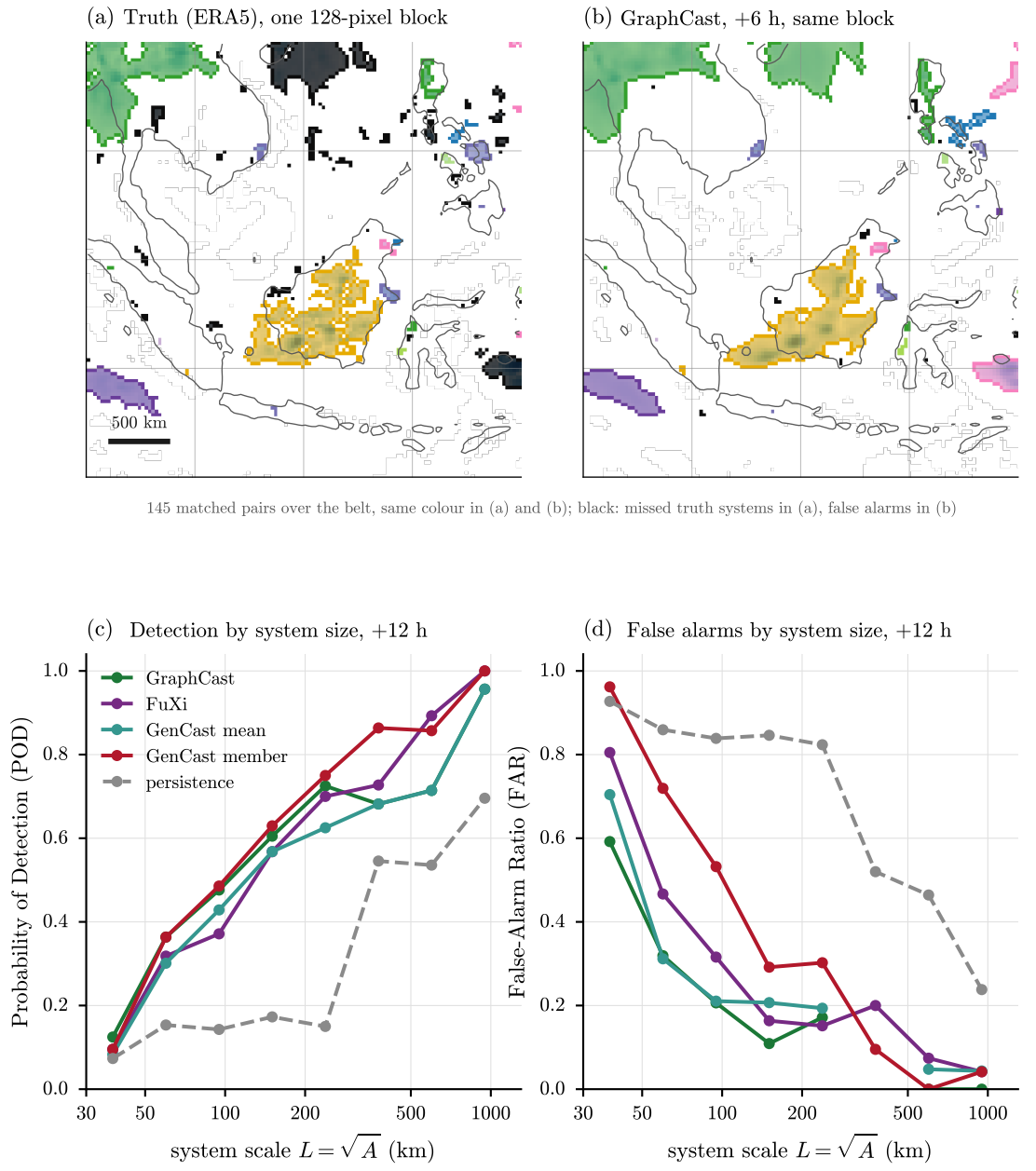

(a) Truth (ERA5), one 128-pixel block
(b) GraphCast, +6 h, same block
500 km
145 matched pairs over the belt, same colour in (a) and (b); black: missed truth systems in (a), false alarms in (b)
(c) Detection by system size, +12 h
(d) False alarms by system size, +12 h
GraphCast
FuXi
GenCast mean
GenCast member
persistence
Probability of Detection (POD)
False-Alarm Ratio (FAR)
system scale $L = \sqrt{A}$ (km)
system scale $L = \sqrt{A}$ (km)
0.0
0.2
0.4
0.6
0.8
1.0
30
50
100
200
500
1000